HANDS ON SUBNETTING THEORY

1.Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

ANS:

Subnetting: Divide 192.168.10.0/24 into 4 Equal Subnets

Given:

- Network address: 192.168.10.0/24
- Number of subnets required: 4
- Borrowed bits: 2
- New CIDR: /26
- Subnet mask: 255.255.255.192

Subnet Table

Subnet	Network Address	Broadcast Address	Subnet Mask
1	192.168.10.0	192.168.10.63	255.255.255.192
2	192.168.10.64	192.168.10.127	255.255.255.192
3	192.168.10.128	192.168.10.191	255.255.255.192
4	192.168.10.192	192.168.10.255	255.255.255.192

Calculation:

- Number of subnets: \(2^2 = 4\)
- Addresses per subnet: \(2^{6} = 64\)
- Usable hosts per subnet: \(64 - 2 = 62\)

**2.For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.**

ANS:

IP Address Assignment for Subnet 1

Selected Subnet: 192.168.10.0/26

- Subnet Mask: 255.255.255.192
- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 – 192.168.10.62

No.	Device	Assigned IP Address
1	Router	192.168.10.1
2	Laptop 1	192.168.10.2
3	Laptop 2	192.168.10.3
4	Printer	192.168.10.4
5	Mobile Phone	192.168.10.5

3.Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

ANS:

Here is a simple subnet plan you can copy into your assignment notebook.

Subnet Plan: 192.168.10.0/24 Divided into 4 Subnets

Main Network: 192.168.10.0/24

Subnet mask: 255.255.255.0

Subnet 1 — /26

Network: 192.168.10.0

Usable: .1 – .62

Broadcast: .63

Router: .1

Laptop 1: .2

Laptop 2: .3

Printer: .4

Mobile: .5

Subnet 2 — /26

Network: 192.168.10.64

Usable: .65 – .126

Broadcast: .127

Other devices

Not assigned

Subnet 3 — /26

Network: 192.168.10.128

Usable: .129 – .190

Broadcast: .191

Other devices

Not assigned

Subnet 4 — /26

Network: 192.168.10.192

Usable: .193 – .254

Broadcast: .255

Other devices

Not assigned

4.Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

ANS:

Step-by-Step Subnetting Calculation

Given Network: 192.168.10.0/24

Step 1: Calculate the New Subnet Mask

We need to divide one network into 4 equal subnets.

- Required subnets = 4
- Borrowed bits = \(2^2 = 4\) subnets
- Original prefix = /24
- New prefix = /24 + 2 = /26

New subnet mask: 255.255.255.192

Step 2: Calculate Hosts per Subnet

- Total bits in IPv4 = 32
- Host bits = \(32 - 26 = 6\)

- Total addresses per subnet = \(2^6 = 64\)
- Usable host addresses = \(64 - 2 = 62\)

We subtract 2 because one address is reserved for the network and one for the broadcast.

Step 3: Find the Subnet Ranges

The block size is \(256 - 192 = 64\). Therefore, the subnet network addresses increase by 64 in the last octet.

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 – 192.168.10.254	192.168.10.255

Step 4: Assign IP Addresses to Devices

I selected Subnet 1, which has valid host addresses from 192.168.10.1 to 192.168.10.62.

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5

All devices have unique IP addresses in the same subnet. The network address (192.168.10.0) and broadcast address (192.168.10.63) are not assigned to devices.

Step 5: Final Summary

- Original network: 192.168.10.0/24
- Number of subnets: 4
- New subnet mask: 255.255.255.192 (/26)
- Total addresses per subnet: 64
- Usable hosts per subnet: 62